Furnace QR scanner set environment (windows)

The technology used Flutter framework, dart (The programming language is designed for client development such as web and mobile apps and can also be used to build server and desktop applications.), and Back4App database.

Step 1: Install the flutter on the local machine. Here is the link to install and straightforward video that will help.

Step 2: Then set your code editor in this example VsCode is used. The small video to help set vs code, and here is the doc. `git clone ["place the URL"]` use any code editor to open the app. Here is an example: VSCODE.

Step 3: create a branch and start working

Step 4: check whether the flutter is set perfectly on this folder, by typing "flutter doctor".

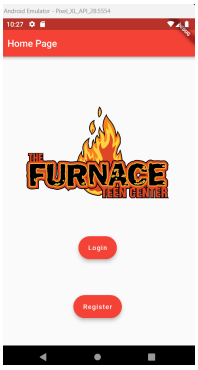
```
[redacted] MINGW64 ~/git/theFurnaceQRAppVscode/furnace-app (anitha)
$ flutter doctor
Doctor summary (to see all details, run flutter doctor -v):
[√] Flutter (Channel stable, 3.7.3, on Microsoft Windows [Version 10.0.22621.1265], locale en-US)
[√] Windows Version (Installed version of Windows is version 10 or higher)
[√] Android toolchain - develop for Android devices (Android SDK version 31.0.0)
[√] Chrome - develop for the web
[√] Visual Studio - develop for Windows (Visual Studio Community 2022 17.0.0 Preview 2.1)
[√] Android Studio (version 2020.3)
[√] IntelliJ IDEA Community Edition (version 2021.2)
[√] VS Code (version 1.75.1)
[√] Connected device (3 available)
[√] HTTP Host Availability

• No issues found!
```

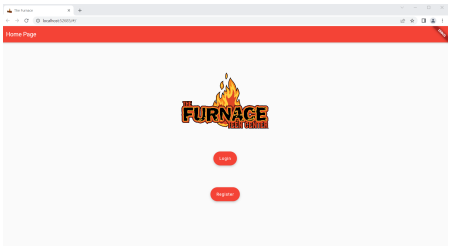
Step 5: Run `"flutter pub get"` which will install all dependence

Step 6: This step can be worked in a different way Android Emulator is best to work.

1. Video to Run Flutter App In VSCode On Android Emulator.

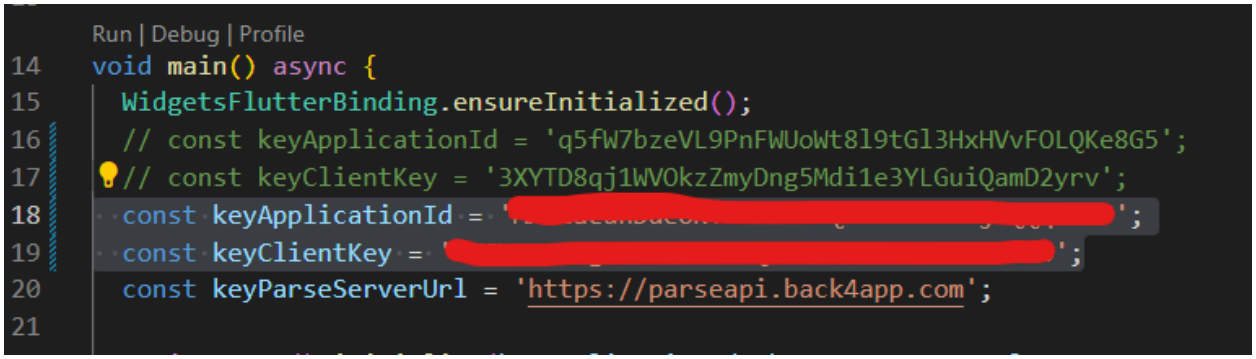


2. Click start debugging for or shortcut "F5" which will start the application.



Set the testing database:

Create an account in back4app, then build the new app in my app, then from the "App Settings" get the key and Id and place them as shown. Make sure at end of the project to replace the old ID and Key. For now, this project does not have a .env file.



```
Run | Debug | Profile
void main() async {
  WidgetsFlutterBinding.ensureInitialized();
  // const keyApplicationId = 'q5fW7bzeVL9PnFWUoWt8l9tGl3HxHVvFOLQKe8G5';
// const keyClientKey = '3XYTD8qj1WVOkzZmyDng5Mdi1e3YLGuiQamD2yrv';
  const keyApplicationId = '[redacted]';
  const keyClientKey = '[redacted]';
  const keyParseServerUrl = 'https://parseapi.back4app.com';

```

Backend/Database

This app creates a backend and database through a website called back4app. This website creates MongoDB databases for applications. It can store any kind of information such as user login info. Back4app can be easily used with Flutter. Examples of how to use CRUD (Create, Read, Update, Delete) applications can be shown here: Flutter CRUD operations or below.

Create:

```
Future<void> saveTodo(String title) async {
    final todo = ParseObject('Todo')..set('title', title)..set('done', false);
    await todo.save();
  }
```

Read:
```
Future<List<ParseObject>> getTodo() async {
    QueryBuilder<ParseObject> queryTodo =
        QueryBuilder<ParseObject>(ParseObject('Todo'));
    final ParseResponse apiResponse = await queryTodo.query();

    if (apiResponse.success && apiResponse.results != null) {
      return apiResponse.results as List<ParseObject>;
    } else {
      return [];
    }
  }
```

Update:
```
Future<void> updateTodo(String id, bool done) async {
    var todo = ParseObject('Todo')
      ..objectId = id
      ..set('done', done);
    await todo.save();
  }
```

Delete:
```
Future<void> deleteTodo(String id) async {
    var todo = ParseObject('Todo')..objectId = id;
    await todo.delete();
  }
```

Barcode Scanner

This app uses a barcode scanner to scan into each room in the teen center. Flutter has special ways it uses a barcode scanner. To learn more about the use of it in Flutter and the full documentation of it click on this website: Flutter barcode scanner. You can also watch this video: Flutter barcode scanner video

Key ideas include:

1. First, install the library: run `'flutter pub add flutter_barcode_scanner'` in your terminal.
2. Importing the library in your class by placing this at the top of the class.

```
import 'package:flutter_barcode_scanner/flutter_barcode_scanner.dart';
```

3. This is an example of calling the method:

```
String barcodeScanRes = await FlutterBarcodeScanner.scanBarcode(
                                        COLOR_CODE,
                                        CANCEL_BUTTON_TEXT,
                                        isShowFlashIcon,
                                        scanMode);
```

4. Here is an explanation of each parameters used in the method from the website:

COLOR_CODE: is hex-color which is the color of the line in the barcode overlay you can pass the color of your choice,

CANCEL_BUTTON_TEXT: is a text of cancel button on the screen you can pass the text of your choice and language,

isShowFlashIcon: is a bool value used to show or hide the flash icon,

scanMode: is an enum in which a user can pass any of { QR, BARCODE, DEFAULT }, if nothing is passed it will consider a default value which will be QR. It shows the graphics overlay like for barcode and QR.